UNIVERSITI TEKNOLOGI MARA
TEST

COURSE	:	**DIGITAL ELECTRONICS**
COURSE CODE	:	**ITT270**
NAME OF COURSE LECTURER	:	
UiTM STUDENT NAME	:	
UiTM STUDENT CARD NO	:	
FACULTY	:	**KPPIM**
PROGRAMME/CODE	:	**CS110/CS143**
PART	:	
GROUP	:	
TIME	:	**2 HOURS**

INSTRUCTIONS TO CANDIDATES

1. Fill in the particulars above.

2. Do not bring any material into the examination room unless permission is given by the invigilator.

3. Answer **ALL** questions

PART	QUESTION	MARKS
A	1 - 10	/20
B	1	/4
	2	/6
	3	/7
	4	/3
	5	/6
	6	/4
	TOTAL	/50

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This test paper consists of 5 printed pages

PART A (20 MARKS)

1. Which of the following statements best describes a combinational logic circuit?
   A. Its output depends only on the current inputs.
   B. Its output depends on both current and past inputs.
   C. It stores data temporarily.
   D. It has memory to store previous inputs.

2. You need to design a circuit that produces a high output when either of two switches is pressed. Which gate would you use?
   A. AND gate
   B. OR gate
   C. NOR gate
   D. NAND gate

3. If a system is designed using the binary number system, what would the output of the logic gate "1101 AND 1011" be?
   A. 1000
   B. 1111
   C. 1001
   D. 1011

4. In a digital circuit, if the output of a 3-input AND gate is 1, which of the following input combinations is possible?
   A. A = 0, B = 1, C = 1
   B. A = 1, B = 1, C = 1
   C. A = 0, B = 0, C = 1
   D. A = 1, B = 0, C = 1

5. Which of the following is the correct output of a 2-input NOR gate when both inputs are 1?
   A. 1
   B. 0
   C. Undefined
   D. 2

6. You are designing a circuit where the output is high only when both inputs are different. Which logic gate should you use?
   A. AND gate
   B. XOR gate
   C. NAND gate
   D. NOR gate

7. How many unique digits are used in the hexadecimal number system?
   A. 7
   B. 8
   C. 15
   D. 16

8. Which of the following best explains why binary is used in digital electronics?
   A. It is easier to read.
   B. It allows faster computations.
   C. Digital systems only understand two states, represented as 0 and 1.
   D. It reduces power consumption in circuits.

9. What is the hexadecimal representation of the binary number 11111110?
   A. FB
   B. 254
   C. 367
   D. FE

10. Using De Morgan's Theorems, how can you express $\overline{\boldsymbol{A}+\boldsymbol{B}}$?
   A. $\overline{\boldsymbol{A}}.\overline{\boldsymbol{B}}$
   B. $\overline{\boldsymbol{A}}.\boldsymbol{B}$
   C. $\overline{\boldsymbol{A}}+\boldsymbol{B}$
   D. $\overline{\boldsymbol{A}+\boldsymbol{B}}$

PART B (30 MARKS)

QUESTION 1

a) What are 2 (TWO) advantages of digital technology?

(2 marks)

b) Why is it important to learn number systems in digital electronics? Provide 2 (TWO) reasons.

(2 marks)

QUESTION 2

Convert the following numbers (show your work):

a) 01111001_{BCD} to hexadecimal

(2 marks)

b) 3572_8 to binary

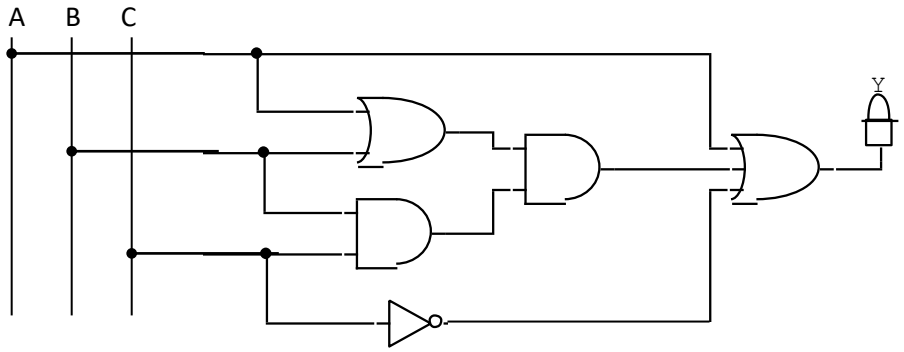
(2 marks)

c) 10110110_2 into decimal.

(2 marks)

QUESTION 3

Given the following logic circuit:



a) Write a Boolean expression.

(2 marks)

b) Derive the truth table.

(3 marks)

c) Write the product-of-sum (POS) expressions for Y

(2 marks)

QUESTION 4

Simplify the following equation using Boolean Theorems OR DeMorgan Theorem:

$$Y = \overline{\mathrm{A} + \mathrm{B}}.\overline{\mathrm{C}} + \overline{\mathrm{A}}.\overline{\mathrm{B}}.\mathrm{C} + \overline{\mathrm{A}}.\mathrm{B}.\overline{\mathrm{C}}$$

(3 marks)

QUESTION 5

Given the following Boolean expression

$$\mathrm{Y} = \left(\overline{\mathrm{A}} + \mathrm{B}\right)\overline{\mathrm{C}} + \mathrm{A}.\overline{\mathrm{B}}.\mathrm{C}$$

Draw the logic circuit using

a) NAND gates only

(3 marks)

b) NOR gates only

(3 marks)

QUESTION 6

A local coffee shop wants to automate their coffee machine's operation based on customer preferences. The coffee machine can make three types of coffee: Espresso (E), Cappuccino (C), and Latte (L). The machine operates under the following conditions:

- Espresso can only be made if the Espresso switch (S1) is ON.
- Cappuccino can be made only if the Cappuccino switch (S2) is ON and the Espresso switch (S1) is OFF.
- Latte can be made if the Latte switch (S3) is ON, regardless of the other switches.

Based on the above information, derive the truth table.

(4 marks)

END OF QUESTION